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Policy Optimization for Pedestrian Traffic Management by Surrogation of Simulation Models

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***Abstract*—This paper proposes a method to swiftly develop crowd control policies that enhance the smooth flow of pedestrian traffic. The method employs a novel, gradient-based black-box optimization technique, which uses a neural network to create a surrogate model replicating a multi-agent simulator. This approach facilitates the rapid generation of improved policies by leveraging gradient information, thereby significantly decreasing the time required to determine optimal strategies. By treating the simulator and its evaluation function as differentiable entities, the method allows for quick policy adjustments guided by gradient information. The proposed method quickly and accurately obtains policy parameters, adapting seamlessly to different optimization strategies in various scenarios. Our methodology was tested by applying it to actual pedestrian flow data from Koshien Stadium in Hyogo, Japan. The real-world application not only confirmed the practical utility of our optimization technique in effectively managing crowd scenarios but also marked a significant advancement in public safety measures for densely populated areas.**

***Index Terms*—Pedestrian Flow Control, Simulation, Machine Learning**

I. Introduction

Public safety is significantly at risk in densely populated places due to many factors. Overcrowding can expedite accidents, obstruct emergency evacuations, and elevate the potential for criminal or terrorist activities. Additionally, events attracting large crowds, like sports fixtures, concerts, and live performances, often exacerbate traffic congestion in nearby areas. Such crowd density can overburden public transportation systems, leading to extreme overcrowding at train and tram stations and bus stops. To address these challenges, the implementation of strategic measures is vital. Effective crowd management, rigorous surveillance, enhanced security protocols, and traffic planning are essential to mitigate congestion and ensure public safety. These proactive steps alleviate immediate risks and foster a safer, more orderly environment for residents and visitors in densely populated areas.

Besides, crowd control through behavioral change is increasingly in focus. For instance, strategies like staged exit procedures for attendees and incentives for staggered departures to transportation hubs can be deployed to manage post-event congestion. This approach helps avoid the rush towards stations and bus stops simultaneously.

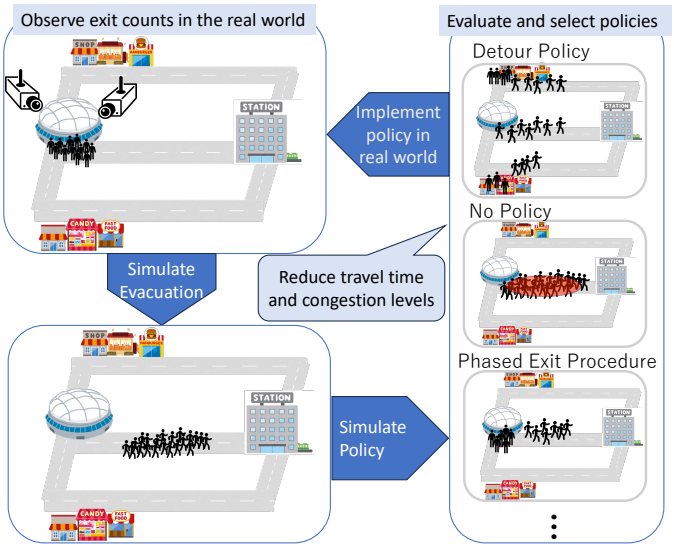


Fig. 1. Policy Optimization via Simulation

However, predicting the effectiveness of policies to manage large crowds, particularly regarding cost efficiency and participant satisfaction, poses a significant challenge. Furthermore, conducting real-world tests to assess the most effective strategy can require substantial investments. This needs a careful balancing act between theoretical planning and practical application, ensuring that the solutions are financially feasible and effective in enhancing public safety and experience. While multi-agent simulations are commonly used for this purpose as shown in Figure 1, the sheer number of policy options and the extensive time required for large-scale simulations make traditional methods such as grid search, random search, and Bayesian optimization inefficient. These methods fail to

leverage real-time pedestrian traffic data, resulting in a significant adaptation shortfall to the dynamic nature of pedestrian movements. This indicates the importance of reducing the time needed to determine the best policies, allowing for utilization of the most precise and timely data.

In this study, we propose a method to swiftly develop control policies aimed at alleviating congestion in large crowds. Our method involves a novel gradient-based, black-box optimization technique. This technique utilizes a neural network to create a surrogate model of a multi-agent simulator, effectively speeding up the policy development process. By leveraging the gradient information, our method enables the generation of improved policies. The concept revolves around replicating the simulator and its evaluation function as differentiable functions. This replication provides insights into how changes in the input could enhance the evaluation, guided by the gradient information. By updating the inputs in accordance with suggested directions of change, we can substantially decrease the number of simulation runs needed to arrive at the optimal policies. This method streamlines the policy optimization process, making it more efficient and effective in handling the complexities of crowd management scenarios.

Our method first establishes a simplified network model that abstracts the return-home scenario from event venues. The simplification facilitates the model's application to various regions while reducing information complexity. After defining this network model, we simulate diverse scenarios to generate a dataset. The dataset forms the foundation for training a surrogate model. The purpose of the surrogate model is to deduce the intricate relationships between various elements like policies, pedestrian flow, geospatial information, and values of the objective function. Once the surrogate model is trained, we utilize it to execute a gradient descent algorithm. The goal of employing gradient descent is to fine-tune policy parameters with the aim of minimizing the objective function. Through this process, we identify optimal policy parameters, which are instrumental in enhancing the overall efficiency and effectiveness of crowd management strategies.

The experimental results from 30 different scenarios demonstrate the efficacy of the proposed method. It achieved an average reduction of 14,507 seconds in exploration time while maintaining comparable scores. Furthermore, our method has achieved accuracy equivalent to that of exploring policies by running the simulator in various scenarios where different optimization policies are applied. Additionally, through evaluations using real-world data, we have observed high performance in terms of computational cost and policy score. This result emphasizes the practical utility and effectiveness of our proposed optimization technique in managing crowd scenarios.

II. Related Work

A. Pedestrian Control

Various studies have been conducted to control pedestrian flow to alleviate congestion during evacuations and reduce evacuation times. Some studies model the environment and constraints as mathematical equations to minimize travel time and evacuation delay. Khalilpourazari et al. [1] propose a robust mathematical model for designing an efficient flood evacuation plan in disasters. Samany et al. [2] propose a method of finding a safe area or shelter using modified particle swarm optimization (MPSO) with local search (LMPSO). Daniele et al. [3] propose a network model that minimizes travel time and evacuation delay. He et al. [4] introduce a bi-level multi-objective location-allocation model based on sequential decision logic to maximize economic sustainability and social utility. These studies suffer from the limitation that they cannot capture the temporal changes in the environment, interactions between agents, and context-dependent decision-making, making them unrealistic.

On the other hand, some studies aim to minimize travel time and evacuation delay in evacuation scenarios using multi-agent simulations, where individuals are treated as agents, and their behavioral rules and interactions are modeled. Dong et al. [5] propose a method combining the cellular automaton model and the differential evolution algorithm to optimize pedestrian flow around large stadiums. Guo et al. [6] propose a simulation-based approach integrating Random Forest (RF) and Non-dominated Sorting Genetic Algorithm III (NSGA-III) to perform evacuation evaluation and optimization at metro stations. Schantz et al. [7] formulate and solve the problem of minimizing the evacuation time of a crowd from a complex building using rescue guides. However, these studies face a challenge due to the time-consuming nature of executing large-scale multi-agent simulations multiple times, as they employ methods such as Bayesian optimization or genetic algorithms to search for optimal parameters.

Several deep learning techniques have been utilized to control pedestrians in more complicated situations since deep learning models can analyze high-dimensional and large datasets in real-time, enabling us to make quick and informed decisions [8], [9]. Fei et al. [10] propose a pedestrian-vehicle mixed-flow model to simulate the parking lot evacuation process at the microscopic level. Wu et al. [11] propose a novel multi-agent recurrent deep deterministic policy gradient (MARDDPG) algorithm based on a deep deterministic policy gradient (DDPG) algorithm for traffic light control (TLC) in vehicular networks. Han et al. [12] propose a pedestrian-considered traffic signal control method using deep reinforcement learning. Zhu et al. [13] propose a context-aware multi-agent broad reinforcement learning (CAMABRL) approach based on broad reinforcement learning (BRL) for mixed pedestrian-vehicle adaptive traffic light control (ATLC). Many of these methods utilize reinforcement learning, which offers the advantage of faster execution as optimal parameters can be obtained in a single run. However, reinforcement learning-based approaches face challenges when applied to scenarios where the optimization objectives vary for each scenario, as once trained, they cannot easily adapt to changes in the objective function.

B. Surrogation-based Optimization

Meanwhile, traditional optimization methods, which often require multiple evaluations of the objective function, can be impractical in such scenarios due to computational cost or time constraints. The concept of surrogate models in optimization emerged from the need to solve complex problems where the objective function is computationally expensive, nonlinear, or even unknown in closed form. Although the primary objective is to speed up the optimization process by replacing the computationally expensive objective function with a surrogate model, the quality of the optimization is tied to how well the surrogate approximates the real objective function.

White et al. [14] proposed a gradient-based nonlinear topology optimization method on a surrogation model for macroscale elastic structures. Kim et al. [15] propose an algorithm for mixed-integer nonlinear simulation-based problems that use adaptive sampling and surrogate modeling. Shirobokov et al. [16] introduce a novel algorithm for optimizing stochastic non-differentiable simulators using deep generative surrogates. Lye et al. [17] present a novel active learning algorithm, termed iterative surrogate model optimization (ISMO), for robust and efficient numerical approximation of PDE-constrained optimization problems. Meng et al. [18] propose an enhanced CO method based on the adaptive surrogate model to achieve high-fidelity multidisciplinary design optimization of the wing. Bui et al. [19] propose an optimization algorithm based on Deep Reinforcement Learning (DRL) for designing power converters. Lim et al. [20] introduce an approach to optimize the Wet Flue Gas Desulfurization (WFGD) system for SOx capture by substituting high-grade limestone with various waste sea shells. Zhang et al. [21] introduce MPFEN, a Convolutional Neural Network (CNN)-based surrogate model designed to enhance the performance of the Multi-Objective Evolutionary Algorithm based on Decomposition (MOEA/D). Seo et al. [22] propose a system that evaluates the quantitative risk in the affected area using a deep neural network surrogate model to determine optimal evacuation routes by integer programming. These methods enable fast and efficient optimization by eliminating the need to run the simulator during optimization. Furthermore, by introducing a differentiable optimization evaluation formula composed of the surrogation model and hyperparameters, rather than directly using the output of the surrogation model as the evaluation value, it becomes possible to achieve optimization in different optimization policies without retraining the model simply by changing the hyperparameters.

C. Our Contribution

The contributions of this paper are summarized as follows. Firstly, it introduces a surrogate model of a multi-agent simulator using a neural network to predict the effectiveness of policies to manage large crowds and proposes a method to optimize policies. Preparing a surrogation model for multi-agent simulators makes it possible to adapt to different optimization strategies for each policy and achieve fast and efficient optimization. Many existing studies rely on Bayesian optimization or reinforcement learning, which often require significant execution time for searching policy parameters and necessitate retraining when optimizing for scenarios with different optimization policies. Secondly, we utilize multi-agent simulation, including policies that involve stochastic behavioral changes in people. Many existing studies use simulators that do not consider human decision-making or path selection, and the policies often do not involve human decision-making interventions that ensure compliance. Furthermore, we conduct evaluations using real pedestrian flow data from the Koshien Stadium in our study. We have observed high performance in terms of computational cost and policy score through evaluations using real-world data.

III. Problem Formulation

In this section, we describe our target scenarios and the problem formulation. Our target scenarios are events such as sports games, concerts, and music festivals where fans or audiences gather at a venue and start leaving the venue at the same time. We assume that the origin of all people's routes is the event venue, and they head for one of the destinations, such as bus stops and stations. We also assume that event managers want to find a *policy* to avoid all pedestrians rushing to the final destinations (e.g., stations) at the same time. The policy consists of a preferable setting of people's drop-by spots (e.g., restaurants) and sojourn time there, as well as the number of stages and the time interval between stages in the phased exit procedure from the venue. The behavioral change to let them drop by or stay at those spots is performed by incentives such as coupons with a monetary cost.

More concretely, a policy consists of two parameters defined for the venue and two parameters defined for each spot. The former two indicate how many stages people should be divided into when exiting the venue and how long people should stay between divided stages. The latter two indicate how strongly each spot should attract people and how long they should stay there. Given a policy, we may simulate how people move according to a reward-response model, route selection model, and other models implemented in a multi-agent simulator, through which the event managers can observe congestion level, travel time of agents to reach the final destination, and the monetary cost to control the agents to fit with the policy.

Given a scenario that represents the road network and the number of people departing from the venue along with their destinations, our objective is to find a policy that minimizes the sum of the congestion level, monetary cost, and travel time. Our objective function $\mathcal{F}$ is defined as the linear combination of a penalty $\mathcal{P}$ for congestion, the amount of the incentive cost $\mathcal{M}$, and the travel time $\mathcal{T}$:

$$\underset{policy}{\operatorname{argmin}}\ \mathcal{F} \tag{1}$$

$$\mathcal{F} = w_1\mathcal{P} + w_2\mathcal{M} + w_3\mathcal{T} \tag{2}$$

where w_1, w_2 and w_3 are weighting factors. Hereafter, we refer to the values output by the objective function as the "policy effectiveness score." Table I lists the notations used in

TABLE I
NOTATION

Symbol	Meaning
N	Set of nodes
n_{origin}	Origin node
D	Set of destination nodes
I	Set of intermediate nodes
E	Set of edges
A	Set of pedestrians
n	Number of participants
d_k	Destination of pedestrian k
$c_{i,t}$	Population of node $i \in N$ at time t
T	Simulation time length
$f_{i,t}$	Maximum population that a transportation facility can handle at destination node $i \in D$ at time t
g_i	Attractiveness of node $i \in N$
s_i	Staying time duration of node $i \in N$
s_{max}	Maximum staying time duration
α	Minimum incentive cost required to gather people
p	The number of stages in the phased exit procedure.
b	Time interval between stages in the phased exit procedure.
b_{min}	Minimum time interval between stages in the phased exit procedure.
b_{max}	Maximum time interval between stages in the phased exit procedure.

this paper. We describe the definitions of each factor in the following sections.

As mentioned, event managers leverage simulation to find the optimal policy that minimizes function $\mathcal{F}$. However, the execution of multi-agent simulation is often computationally intensive, which means solution search takes time, even though some heuristics are employed. In our proposed approach described in Section IV, drawing on the teacher-student paradigm, we use deep learning to build a surrogate model for pedestrian traffic simulation. We then employ gradient-based methods, like Gradient Descent (GD), to directly search for the optimal policy parameters.

A. Scenario: Road Network and Travels of People

A road network G of a scenario is represented by set N of nodes and set $E \subseteq N \times N$ of edges. The edges represent roads connecting the nodes. The nodes consist of a single venue, destinations such as stations, and intermediate nodes representing several spots such as commercial facilities. Namely, a node-set consists of three types of nodes, *i.e.*, origin node n_{origin}, a set I of intermediate nodes, and a set D of destination nodes. We define a whole node set N as the union of these three sets $N = \{n_{origin}\} \cup I \cup D$ where, $D \neq \emptyset$, $I \neq \emptyset$, $n_{origin} \neq \emptyset$, $n_{origin} \notin D$, $n_{origin} \notin I$ and $I \cap D = \emptyset$.

The origin node n_{origin} represents the event venue, while the intermediate nodes I encompass intersections and commercial facilities. Meanwhile, the destination nodes D correspond to transportation hubs such as stations and bus stops.

We denote the number of pedestrians (i.e., participants) departing from the event venue as $n \in \mathbb{N}$. While each pedestrian p has a preferred transportation hub $d_p \in D$ and policies can influence their routes, we assume that their destinations are fixed and cannot be altered.

B. Policy

In multi-agent pedestrian traffic simulations assessing the efficacy of a policy, the attractiveness parameter g_i (how pedestrians should prefer node i) is normalized by the total number of nodes, representing the probability that each participant detours to that point. Conversely, the stay-time parameter s_i indicates the duration participants stay at the node i. The stay-time parameter s_i is defined within the range [0,s_{max}]. We randomly select the duration for which each pedestrian stays at the node i from a uniform distribution with a range [0, s_i]. Furthermore, the parameter p represents the number of stages in the phased exit procedure, indicating how many stages pedestrians are divided into during the exit process, with each pedestrian randomly assigned to a stage. Additionally, the time interval parameter b represents the time interval between departures at each stage in the phased exit procedure. The stay-time parameter b is defined within the range [b_{min}, b_{max}], where the departure time interval between each stage is represented by b.

C. Penalty for Congestion

Regarding congestion levels, while participants may exit the venue soon after its conclusion, accidents threaten the event's sustainability. As crowd density rises, the risk of accidents escalates, and excessive congestion at stations or roads can lead to potential stampedes. Moreover, excessive congestion can cause stress for event participants, diminishing their overall satisfaction. Therefore, we define the penalty $\mathcal{P}$ for congestion, focusing on the congestion at the destinations as below.

$$\mathcal{P} = \sum_{t=0}^{T} \sum_{i \in D} \max(0, c_{i,t} - f_{i,t}) \tag{3}$$

In the above equation, $c_{i,t}$ represents the total number of people who arrived at destination $i \in D$ by time t. Note that we assume people start leaving the venue at time 0. Given that the number of people at the destination changes due to the arrivals and departures of transports, we subtract the maximum capacity $f_{i,t}$ of the transport at destination i at time t from $c_{i,t}$ and accumulate the number of remained people over time.

D. Incentive Cost

No matter how safe or quickly people can return home, if too much of a cost is spent on the policy, it will be impossible to alleviate congestion continuously. Therefore, the monetary cost $\mathcal{M}$ is defined as below.

$$\mathcal{M} = p\frac{b}{b_{max} - b_{min}} + \sum_{i \in I}(\frac{s_i}{s_{max}} + \alpha)g_i \tag{4}$$

In the above equation, α represents the minimum incentive cost required to attract pedestrians, and g_i represents the attractiveness of node i. Since encouraging people to stay longer generally incurs more monetary costs, we also consider the stay duration parameter s_i. p denotes the number of divisions in the phased exit, and b represents the time interval between each division.

E. Travel Time

Lastly, regarding travel time, even if we can manage pedestrians reaching their destinations with minimal congestion and at a low monetary cost, the longer it takes, the more it adversely affects the satisfaction of event participants. Therefore, we define the travel time cost $\mathcal{T}$ as follows:

$$\mathcal{T} = \frac{\sum_{t=1}^{T} \sum_{i \in D} (c_{i,t} - c_{i,t-1}) t}{n} \tag{5}$$

The above definition accumulates the travel time for each pedestrian from the origin to their destination.

IV. Proposed Method

For policy evaluations and optimization, agent-based pedestrian simulations are commonly utilized. These simulations provide a detailed perspective, yet they are computationally expensive and time-consuming, making real-time policy optimization impractical.

To address these challenges, we propose two key approaches. Firstly, we introduce a deep learning-based surrogate model trained on traffic simulation, rapidly estimating the influence of diverse policies. Since deep learning models are inherently differentiable, it allows for gradient-based optimization techniques, such as gradient descent, to be employed directly. Secondly, we utilize a simplified virtual road network named detour-centric road network. This focuses on a core structure composed of a single "origin node," multiple "destination nodes," and multiple "intermediate nodes." The purpose of this simplification is to abstract the complexities of real-world road networks into a tractable model for policy evaluations. The overview of the training surrogate model is shown in Figure 2.

A. Detour-Centric Road Network

In order to efficiently evaluate the effects of policies, we introduce a detour-centric road network for simulations, a simplified virtual road network tailored for agent-based traffic simulations. This network is specifically designed to address the challenge of prolonged simulation execution times and the complexity of learning that arise when directly inputting the entire real-world road network.

The detour-centric road network comprises only essential nodes (i.e, origin node, intermediate nodes and destination nodes) and edges, as shown in Figure 3. The goal is to simulate the effects of policies that may guide agents to detour through these nodes before reaching their final destination. The edges connect the origin node to each intermediate node and from each intermediate node to the destination node. Note that edges directly connect the origin node and each destination node, functioning as bypass routes and serving as the shortest paths. We also note the weight of each edge mirrors the average travel distance within the actual city.

By trimming the excess intricacies of real-world road networks and focusing solely on pivotal nodes, the detour-centric road network dramatically cuts down simulation run times and facilitates emulation through learning. Moreover, the network is optimized for simulating the effects of policies, thereby ensuring that policy evaluations are both effective and representative of potential real-world implementations.

B. Training Surrogate Model

As depicted in Figure 2, this trainable neural network model learns to mimic the output of the agent-based traffic simulation. A significant benefit of constructing the surrogate model using deep learning is that the entire model becomes differentiable. This differentiability paves the way for gradient-based optimization methods such as gradient descent. Using gradient information, we can efficiently search for policies that minimize the policy effectiveness score, thus identifying the optimal policy more rapidly than the traditional simulation-based method.

The inputs to the model consist of the policies to be verified, topography information, and agent behavior. However, directly inputting the behavior of all pedestrians into the model, as in a simulator, is challenging. Therefore, we convert agent behavior into values representing the proportion of pedestrians and the utilized destination nodes and input these values instead. The model output consists of penalty for congestion and travel time, constituting the policy effectiveness score. On the other hand, incentive cost is defined by a differentiable expression derived from policy parameters thus model outputs incentive cost directly as the expression. Rather than directly predicting the policy effectiveness score, predicting each component separately improves prediction accuracy and allows for optimization with different weights for each component to minimize them.

C. Policy Optimization Using Surrogate Model

We leverage the trained surrogate model to search for optimal policy parameters. Using the gradient information of the model, we iteratively refine the policy parameters to minimize the objective function based on the gradient descent. Considering our problem as a constrained optimization task, policies have predefined bounds, ensuring stay times remain non-negative and within a specific range. To maintain these constraints while optimizing, we apply Projected Gradient Descent (PGD). PGD ensures that if the solution obtained after applying gradient descent at each step is outside the specified constraint range, it is projected onto the nearest boundary of the constraint range. This ensures that the solution always adheres to the predefined constraints.

Following only the gradients can lead to a local optimal solution. To mitigate this, we initialize the optimization process with parameters set in a grid fashion. Out of various solutions obtained from different initial values, the one yielding the best-estimated score in the surrogate model is considered as the optimal policy. This optimization process is shown in Algorithm 1.

V. Experiments and Results

This section explains experimental conditions and results. The objective of the proposed method is to derive optimal policies swiftly while accommodating variations in optimization

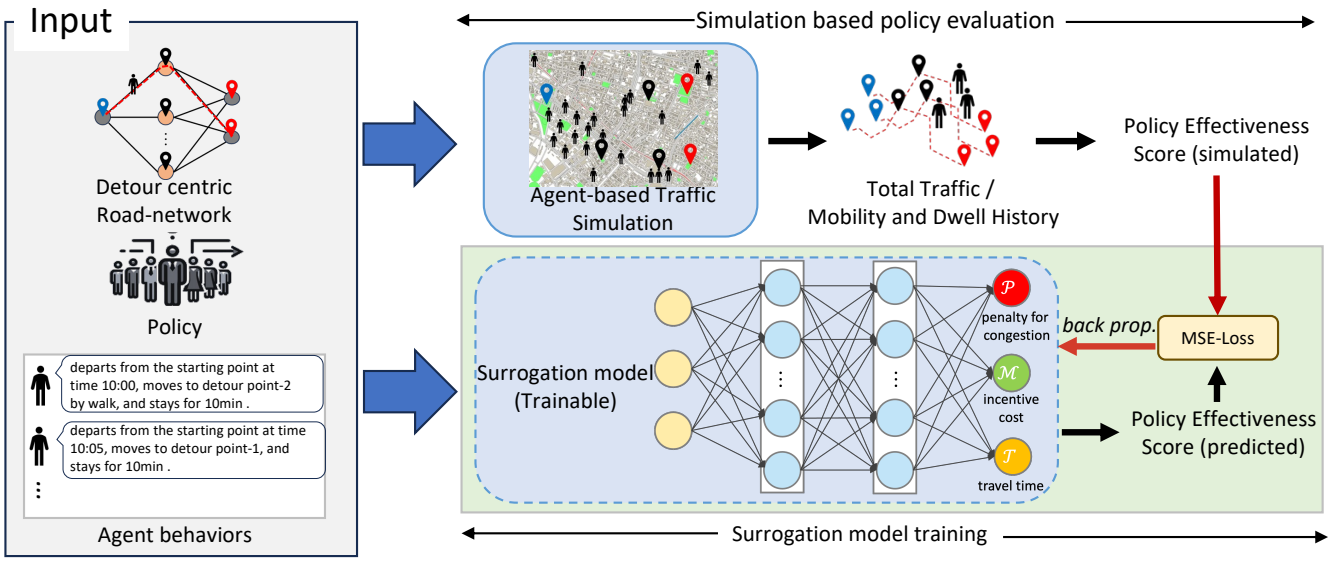


Fig. 2. Training of Surrogate Model

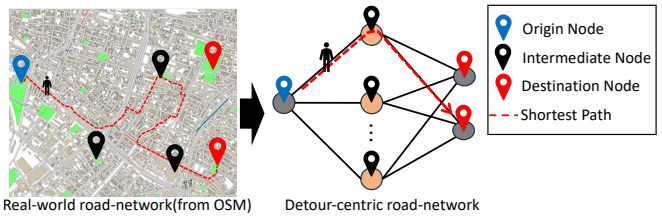


Fig. 3. Converting OSM (Open Street Map) Road Network to Detour-Centric Road Network

strategies across scenarios. Therefore, the evaluation focused on three aspects: the processing time required compared to other methods, the quality of policies relative to other methods considering processing time, and the quality of policies when varying the optimization strategy (weights of each element of the policy effectiveness score).

A. Evaluation Environment and Scenario Setting

For pedestrian flow simulation, we utilize Space-Time Engineering's Scenargie® software. We prepared scenarios with randomly set geographical and pedestrian flow information to enable learning and validation in various geographical and pedestrian flow scenarios. Additionally, we prepared scenarios based on real data to verify the effectiveness of the proposed method in actual environments.

1) Randomly Genarated Virtual Scenario: In this experiment, we created scenarios with randomly set road networks and pedestrian flow conditions to evaluate whether optimization can be achieved under different road networks and pedestrian flow conditions. In our simulation, we designed a $2km \times 2km$ field that reflects real-world scenarios, encompassing the nearest stations, venues, and commercial facilities based on actual events. For the scenarios used in the experiments, we randomly placed stations, commercial facilities, and event venues within this field and generated random road networks that all agents must traverse. The venue to be deployed is 1, and the number of target commercial facilities for policies and stations is randomly set from 1 to 3. Drawing from real-world train data, we configured the stations' capacity f to handle 1,500 people every 3 minutes. The simulation time length T was set to 7,200 seconds, allowing sufficient time for all pedestrians to exit and for the processing

Algorithm 1 Grid-Initialized Policy Optimization

Input: grid of parameters $\boldsymbol{\Theta}$ (sampled from the policy parameter space), learning rate α, convergence criterion ϵ, surrogate model $\tilde{\mathcal{F}}$, detour-centric road network $\mathcal{G}$, agent behavior $\mathcal{B}$
Initialization
optimal parameters: $\boldsymbol{\theta}^* \leftarrow$ randomly from $\boldsymbol{\Theta}$
optimal policy effectiveness score: $E^* \leftarrow \infty$
for each parameter set $\boldsymbol{\theta_k} \in \boldsymbol{\Theta}$ **do**
  repeat
    Compute gradient: $\nabla\tilde{\mathcal{F}}(\boldsymbol{\theta_k}, \mathcal{G}, \mathcal{B})$ by back-propagating through $\tilde{\mathcal{F}}$
    Update parameters: $\hat{\boldsymbol{\theta_k}} \leftarrow \boldsymbol{\theta_k} - \alpha\nabla\tilde{\mathcal{F}}(\boldsymbol{\theta_k}, \mathcal{G}, \mathcal{B})$
    Project $\hat{\boldsymbol{\theta_k}}$ onto feasible range: $\boldsymbol{\theta_k} \leftarrow \text{proj}(\hat{\boldsymbol{\theta_k}})$
  until $\|\nabla\tilde{\mathcal{F}}(\boldsymbol{\theta_k}, \mathcal{G}, \mathcal{B})\| < \epsilon$
  if $\tilde{\mathcal{F}}(\boldsymbol{\theta_k}, \mathcal{G}, \mathcal{B}) < E^*$ **then**
    Update optimal parameters: $\boldsymbol{\theta}^* \leftarrow \boldsymbol{\theta_k}$, optimal loss: $E^* \leftarrow \tilde{\mathcal{F}}(\boldsymbol{\theta_k}, \mathcal{G}, \mathcal{B})$
  end if
end for
Output: optimal parameters $\boldsymbol{\theta}^*$

of pedestrians remaining at the stations. Additionally, for the setting of evacuees, the number of participants n is randomly set from 10,000 to 30,000, and the destination selected by each participants is randomly determined. Furthermore, s_{max}, b_{min}, and b_{min} are set for 1200 seconds, 300 seconds, and 600 seconds, respectively. The number of stages in the phased exit procedure p is set from 1 to 5.

2) Scenario Based on Real Data: This dataset is constructed based on pedestrian flow data observed around the Koshien Baseball Stadium in Hyogo before and after baseball games. Pedestrian flow data is regularly aggregated at the regional mesh level using location information obtained from base stations. These data include the movement of pedestrians between meshes and population data for each mesh, with anonymization applied. In this study, pedestrian movements are replicated to maintain consistency in pedestrian flow, and the number of people in the scenario and their destinations are generated from these flows. The number of nodes, roles of nodes, their assignments, and inter-node distances are set based on the respective area's Open Street Map (OSM) data.

B. Representational Fidelity of Detour-Centric Road Network

TABLE II
REPRESENTATIONAL FIDELITY OF DETOUR-CENTRIC ROAD NETWORK

Metrics	Penalty for Congestion	Travel Time
RMSE	0.054	0.11
MSE	0.0029	0.011
MAE	0.022	0.077

First, we assess how much the detour-centric road network represents the original network. For this purpose, we created 100 scenarios and conducted simulations on randomly generated road networks and the detour-centric road network derived from them. Travel time and penalty for congestion are normalized to a scale of 0 to 1. Table II presents the RMSE, MSE, and MAE of each element of the score measured on the abstracted network compared to the ground truth elements measured on the original network. From Table II, it is evident that there are errors in both Travel Time and Penalty for Congestion. To demonstrate that these errors are within acceptable limits, we will examine the performance of the optimization process in the following sections.

C. Accuracy of Surrogate Model

TABLE III
ACCURACY OF SURROGATE MODELS FOR EACH NETWORK CONFIGURATION

Metrics	FCN	GATConv
RMSE	**0.038**	0.043
MSE	**0.0015**	0.018
MAE	**0.028**	0.030
MAPE	**6.1%**	6.4%

Secondly, we validate the estimation accuracy of the surrogate model for simulations on detour-centric road networks while examining which network configuration is optimal. We prepared 10,000 random scenarios for the experiment, creating detour-centric road networks based on each scenario's original road network. We then split the dataset into an 80:20 ratio for training and validation. In this case, we aim to estimate the simple sum of Travel Time and Penalty for Congestion when conducting simulations on a detour-centric road network. Travel time and penalty for congestion are normalized to a scale of 0 to 1. Table III shows the estimation accuracy for various network architectures. Table III shows that networks composed of fully connected layers achieve comparable accuracy to those composed of Graph Attention Networks (GAT). This may be attributed to the fact that the road network, through detour-centric road network representation, has become sufficiently homogenized in geographical differences across scenarios, allowing even fully connected layers to learn effectively.

D. Optimization Result

Thirdly, we investigate whether the policies obtained on the detour-centric road network are effective on the original road network and examine the relationship between policy effectiveness score and execution time. Additionally, we examine whether the proposed method can explore policies in scenarios on which the surrogate model has not been trained. In this validation, we use the FCN model learned in Section V-C to conduct policy exploration on the detour-centric road network and verify the obtained policies by conducting simulations on the original network afterward. For this validation, we prepared 20 original network scenarios that were not used for training the model and the corresponding detour-centric road networks for each of these scenarios. As a comparative method, we utilized Bayesian Optimization and explored optimal parameters in the original network using the simulator, validating the acquired policies by simulating them on the original network. The exploration in Bayesian Optimization was terminated after 50 simulation iterations. The weights of each element constituting the policy score are fixed to assess whether the proposed method can rapidly and efficiently explore policies. Regarding the weights of each element in the policy effective cost, when we evenly distributed all weights, the optimal solution always turned out to be not implementing any policies. Therefore, we set the weights for station congestion, travel time, and policy cost at 1:1:0.1, respectively.

Figure 4 and 5 show the policy effectiveness scores and the execution times at each iteration of Simulation. Bayesian Optimization explores policies through simulation execution, causing variations in scores and execution times with each iteration. In contrast, the proposed method does not require simulation execution during the exploration phase, resulting in constant scores and execution times regardless of the number of iterations. From Figure 4, the proposed method achieves an average policy effectiveness score of 0.38, while Bayesian Optimization achieves an average score of 0.35. This is because the proposed method conducts two-stage

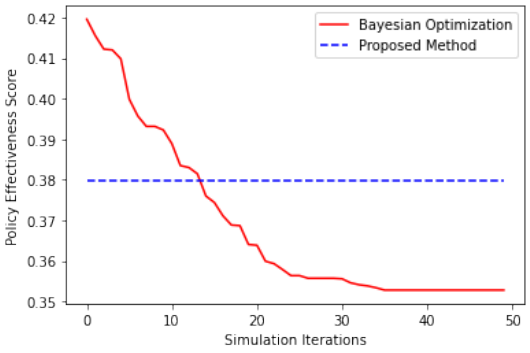


Fig. 4. Average Policy Effectiveness Score in Virtual Scenario

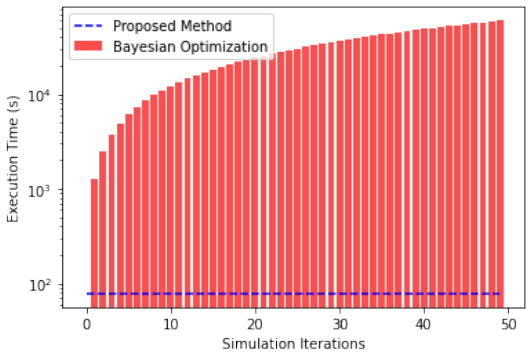


Fig. 5. Average Execution Time in Virtual Scenario

imitation by abstracting the road network and using a neural network to mimic the simulator. Therefore, even if the gradient information suggests an optimal policy, it may not be optimal on the original network during simulation. Hence, to improve optimization performance, it is necessary to develop better configurations of neural networks that mimic a more accurate representation of the road network and simulator.

From Figure 4, it can be observed that Bayesian Optimization requires an average of 12 simulation runs to achieve scores equal to or lower than those obtained by the proposed method. Furthermore, as shown in Figure 5, the average time for 12 simulation runs is 14,585 seconds, whereas the proposed method, which does not require simulation runs for exploration, takes only 78 seconds. Consequently, the proposed method can efficiently search valid policies much faster, reducing policy search time by 14,507 seconds. This reduction in exploration time enables rapid adaptation to changing conditions during event evacuations and allows for comparative analysis of Pareto solutions.

E. Verification of Generality to Optimization Strategy

Fourthly, to investigate whether optimization can be performed even when the weights w_1, w_2, and w_3 constituting the policy cost are changed, we prepared scenarios with different weightings for the ten costs. The weights of each element are set to random values ranging from 0 to 10 for w_1 and w_2, and from 0 to 1 for w_3, as setting the incentive cost too high may always lead to the optimal policy not to take any action. Table IV shows the policy scores for each method in each scenario. From Table IV, it can be confirmed that the proposed method can find better-scored policies in half of the scenarios out of ten, indicating that the proposed method can optimize even with different optimization strategies.

TABLE IV
POLICY EFFECTIVENESS SCORE OF POLICIES WITH DIFFERENT WEIGHTS

Scenario ID	Proposed Method	Bayesian Optimization
1	5.5	**4.9**
2	**1.1**	1.3
3	0.31	**0.30**
4	**1.3**	1.7
5	**3.3**	3.5
6	2.2	**1.6**
7	**1.4**	1.4
8	1.1	**1.1**
9	**1.3**	1.3
10	5.8	**4.9**
average	2.3	**2.2**

F. Verification in Scenarios Based on Real Data

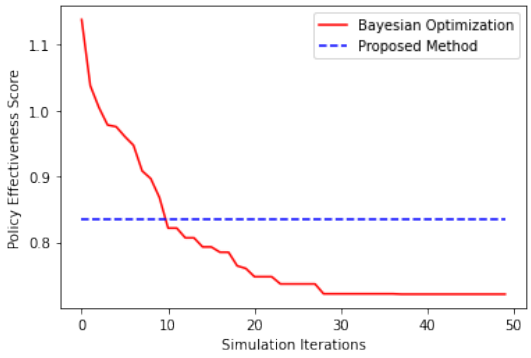


Fig. 6. Average Policy Effectiveness Score in Real Scenario

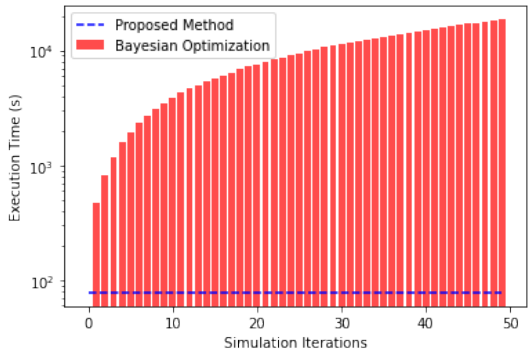


Fig. 7. Average Execution Time in Real Scenario

Finally, to validate the effectiveness of the proposed method in real environments, optimization was performed using scenarios based on actual exit pedestrian flow information and

terrain information, as described in Section V-A2. In this evaluation, the weights of each element were randomly set to values ranging from 0 to 10 for w_1 and w_2 and from 0 to 1 for w_3, and five scenarios were prepared. Figure 6 shows the policy effectiveness score at each iteration of simulation runs, while Figure 7 illustrates the execution time at each iteration. Figure 6 shows that existing methods require ten simulation runs to achieve the same policy effectiveness score as the proposed method. Additionally, from Figure 7, it can be observed that while ten simulation runs take 3,884 seconds, the execution of the proposed method takes only 78 seconds, reducing exploration time by 3,806 seconds. Although the reduction in execution time is smaller compared to randomly generated data, this is attributed to the fact that the data targeted in this study pertains to narrow areas near stations and venues. Nevertheless, the time reduction is still considered effective, demonstrating the validity of the approach.

VI. Conclusion

In this paper, we propose a method for rapidly exploring crowd control policies for large-scale crowds through surrogating agent-based simulators. In the evaluation experiments conducted with 30 scenarios, the proposed method achieved an average reduction of 14,507 seconds in exploration time while achieving comparable scores. Furthermore, we have confirmed that our proposed method achieves equivalent accuracy to exploring policies while running the simulator in scenarios where different optimization policies are applied for each scenario. Additionally, through evaluations using real-world data, we have observed high performance in terms of computational cost and policy score.

We plan to enhance optimization accuracy by exploring better representations for road abstraction and refining the configuration of neural networks emulating simulations. Additionally, we are considering expanding the range of policy options by increasing train frequency and implementing policies that involve walking one station's distance. Moreover, we plan to validate our proposed method in more complex scenarios, incorporating train stations and bus stops as part of the transportation infrastructure. We also consider comparing our method with other state-of-the-art methods, such as reinforcement learning and differential evolution.

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